



University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

Second Year Examination - Semester I - UCSC AY22 [held in September/October 2025]

IS2205 - Mobile Application Design and Development

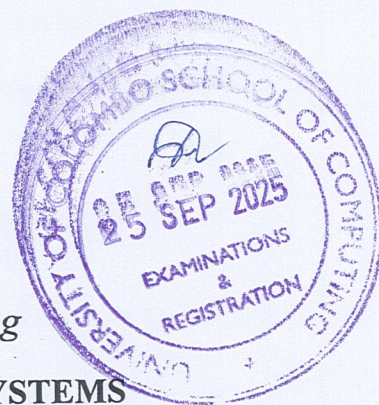
(Two (2) Hours)

Answer ALL questions

Number of Pages = 10

Number of Questions = 4

114



To be completed by the candidate								
Index Number								

Important Instructions to candidates:

- I. Students should answer in the medium of English language **only** using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 10 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

1	
2	
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Total	

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Q1.

- (a) Related to the Android API framework, briefly explain the difference between a sensor and an actuator

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[04 marks]

- (b) Briefly explain a key difference(01) and a similarity(01) between a software based sensor and a hardware based sensor.

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[04 marks]

- (c) Fill the following two(02) blanks to enforce the availability of the accelerometer in the device when installing the application.

```
<uses-feature
android:name="android.hardware.sensor.accelerometer"

    android:_____ = _____ />
```

[04 marks]

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- Native Applications _____ Web Applications
- Web Applications _____ Hybrid Applications

(e) Fill the three(03) blanks such that the core idea of Gestalt priciples of cognition is correctly stated:

The _____ is

other than the _____ of

the _____.'

(f) Briefly define the users' mental functions of emotion and perception, that the UX designer must consider during the design of a mobile application.

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Q2.

(a) Briefly explain following three(03) concepts of the Jetpack library.

- (i) Modifier
- (ii) Composable
- (iii) Preview annotation

[06 marks]

(b) Briefly explain the Kotlin code lines marked with (i),(ii) and (iii) below, via suitable code comments.

```
fun main(args : Array<String>) { //(i)
    for (i in args){ //(ii)
        println(`I: $i`) //(iii)
    }
}
```

(i)

(ii)

(iii)

[07 marks]

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- (c) State two(02) difficulties arise when a mobile application developer tries to use SQLiteHelper to directly interact with SQLite database.

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[04 marks]

- (d) Fill the following code fragment with appropriate words for a Room library based interface definition for an Entity called Student.

```
-----  
interface StudentDao {  
  
    -----(`----- * FROM students')  
  
    fun getAll(): Array<----->  
  
    //Remaining code
```

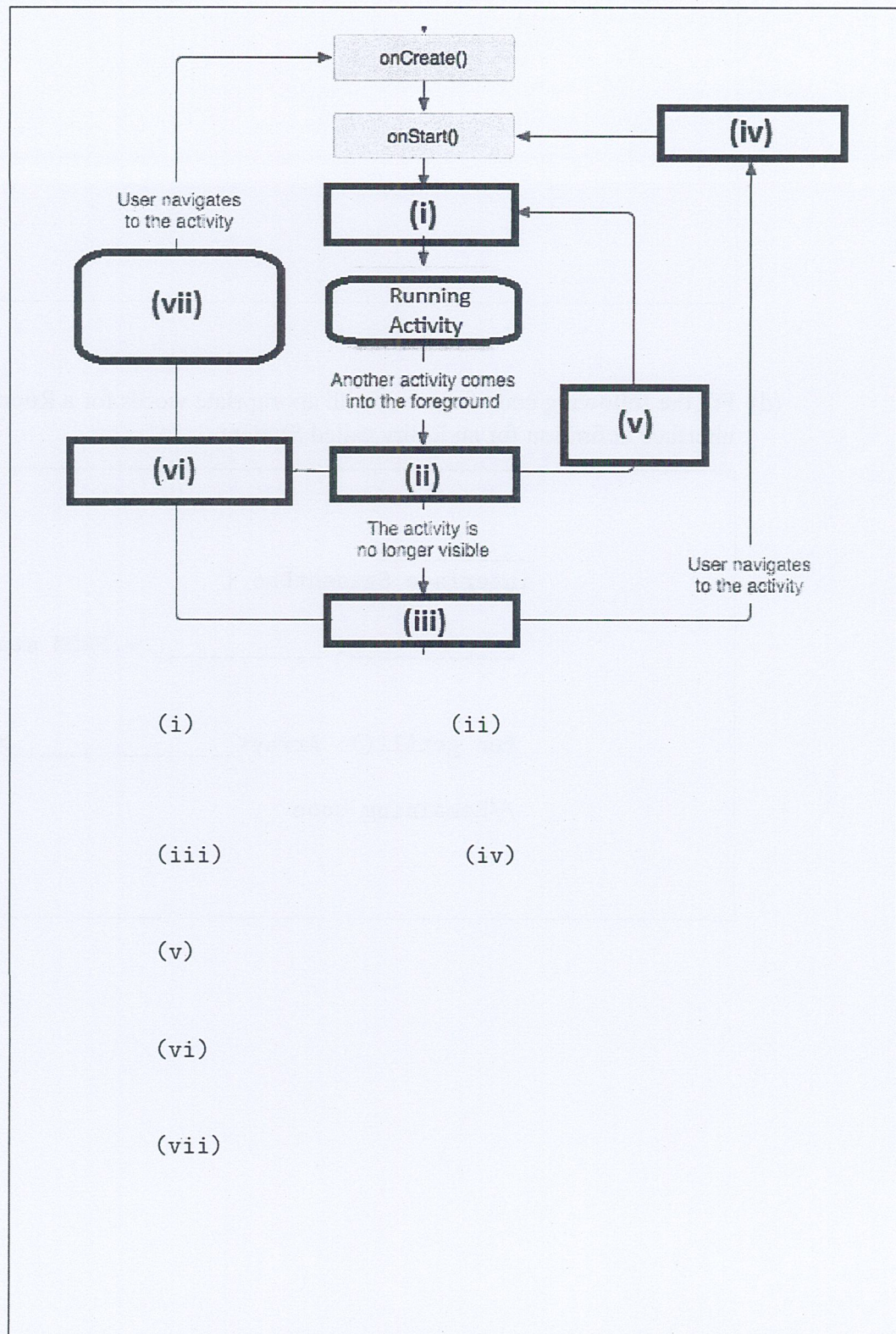
[08 marks]

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Q3.

- (a) In the Android Activity life cycle diagram below (i) to (iv) represent callbacks. (v) to (vii) represent events/states. State the correct callbacks and events/states for the Roman numbers.



[07 marks]

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- (b) Briefly explain the following concepts related to an Android native application.
- (i) Tasks
 - (ii) Backstack

--

[04 marks]

- (c) Briefly explain the following methods related to an Android native application.
- (i) `getCallingActivity()`
 - (ii) `finish()`
 - (iii) `onSaveInstanceState(Bundle)`
 - (iv) `onRestoreInstanceState(Bundle)`

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[08 marks]

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- (d) State the mandatory parameter for declaring a service in an Android manifest file.

--

[02 marks]

- (e) There is a restriction introduced for background services since Android 8.0. State two(02) reasons that can justify the introduction of such a restriction.

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[04 marks]

Q4.

- (a) Refactoring is an important functionality of mobile application development tools. In this context, define the meaning **and** write one(01) concrete example for refactoring.

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[03 marks]

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(b) Calculate the result of the following operations in Dart programming language.

(i) $43 \sim / 7$

(ii) $40 \% 6$

[04 marks]

(c) Briefly explain the purpose of following directory and files of a Flutter project.

(i) lib directory

(ii) analysis_options.yaml file

(iii) pubspec.yaml file

--

[06 marks]

(d) Complete the two(02) blanks in the following code fragment of Dart on **enum** the definition.

```
enum DataSource {  
    one(0),  
    two(1);  
    final int num;  
  
    const _____(_____ .num);  
}
```

[04 marks]

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- (e) Briefly explain the importance of Widget in Flutter framework and state the key difference between the **StatefulWidget** and the **StatelessWidget**.

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[04 marks]

- (f) Briefly explain the how can we acheive the following in a Flutter app, when the sqlite Dart package is used.

- (i) Restricting the table creation SQL statement from executing more than once.
- (ii) Restricting multiple instances of database table file from opening.

--

[04 marks]